

Legal Mate

AI Legal Case Similarity Search

Architecture, Tech Stack & Future Scope

POC Documentation | Prepared for internal review

1. What is Legal Mate?

Legal Mate is an AI-powered semantic search engine for Indian court judgements. A lawyer uploads a case draft (PDF or plain text), and the system automatically:

1. Understands the legal issues in the case
2. Searches live Indian legal databases (Indian Kanoon, LiveLaw, Bar & Bench, SCI)
3. Finds the most semantically similar past judgements
4. Returns structured citations with relevance scores, legal reasoning, and source links

Simple analogy: Think of it as a 'Google for your case' -- but instead of returning web pages, it returns similar court judgements with AI-generated similarity explanations.

2. System Architecture

Two-tier deployment: FastAPI backend (Render) + Streamlit frontend (Streamlit Cloud)

```
USER (Browser)
  |
  | Upload PDF / Paste text
  v
STREAMLIT FRONTEND (legalmate.streamlit.app)
  |
  | POST /search
  v
FASTAPI BACKEND (legal-search-api.onrender.com)
  |
  |-- Step 1: EXTRACTOR -> Gemini reads case, extracts statutes + queries
  |-- Step 2: SEARCHER -> Serper searches 4 legal sites in parallel
  |-- Step 3: SCRAPER -> Fetches full case text from URLs
  |-- Step 4: CHUNKER -> Splits docs into 512-char chunks
  |-- Step 5: EMBEDDER -> Converts chunks to 768-dim vectors
  |-- Step 6: FAISS -> Finds top-20 most similar chunks
  |-- Step 7: VALIDATOR -> Gemini rates legal similarity, returns citations
  |
  v
JSON response -> Streamlit renders cards with similarity scores
```

3. How Similar Cases Are Retrieved (7-Step Pipeline)

Step 1: EXTRACT [gemini-2.5-flash]

Case text/PDF -> JSON with case_type, statutes, keywords, 3 search queries.

Analogy: A trained lawyer reads your case and writes a research brief.

Step 2: SEARCH [Serper.dev (Google Search API)]

3 search queries run on 4 sites in parallel:

- site:indiankanoon.org (main legal database)
- site:livelaw.in (recent HC/SC judgements)
- site:barandbench.com (legal news + judgements)
- site:sci.gov.in (Supreme Court official)

Returns: up to 10 unique case URLs + titles + snippets

Step 3: SCRAPE [Direct HTTP + Serper snippets]

Fetches full case text from each URL.

Fallback: Uses Google snippets when direct access is blocked (datacenter IPs).

Step 4: CHUNK [LangChain RecursiveCharacterTextSplitter]

Splits long judgements (50,000+ chars) into 512-char chunks with 64-char overlap.

Analogy: Breaking a book into readable paragraphs. Max 20 chunks (free-tier quota).

Step 5: EMBED [gemini-embedding-001]

Converts all text chunks + user query into 768-dimensional float vectors.

Analogy: Translating text into a language of numbers where similar meaning = similar numbers.

Batched: All chunks in ONE API call (respects 100 req/min free-tier limit).

Step 6: VECTOR SEARCH [FAISS IndexFlatIP]

Cosine similarity search: query vector vs all chunk vectors.

Returns top-20 most similar chunks in milliseconds.

Analogy: GPS finding nearest locations -- but for legal meaning, not geography.

Step 7: VALIDATE [gemini-2.5-flash]

LLM reads user case + top chunks, rates genuine legal similarity (0.0-1.0).

Returns: case_name, citation, relevance_score, key_overlap, reasoning, source.

Thinking disabled (thinking_budget=0) for speed -- saves 10-15s per call.

4. Tech Stack -- Current vs Alternatives vs Enterprise

Component	Current (POC)	Why Chosen	Enterprise Alternative
LLM	gemini-2.5-flash	Free tier, PDF native, JSON mode	GPT-4o / Claude 3.5 Sonnet
Embeddings	gemini-embedding-001	Same API key, 768-dim, batched	text-embedding-3-large (OpenAI)
Vector DB	FAISS (in-memory)	No DB needed, fast, ephemeral	Pinecone / Qdrant / pgvector
Search	Serper.dev	2500 free/month, works on cloud	IK API + SCC Online + Manupatra
Scraper	Direct HTTP + snippets	Free, no setup needed	Playwright + Firecrawl
Chunker	LangChain text splitter	Battle-tested, easy config	Semantic chunker
Backend	FastAPI + Uvicorn	Async, auto-docs, fast	FastAPI + Kubernetes
Frontend	Streamlit	POC speed, 1-file UI	React + Next.js
Backend Host	Render (free)	GitHub auto-deploy, 0 cost	AWS ECS / GCP Cloud Run
Frontend Host	Streamlit Cloud	Free, shareable URL	Vercel / AWS CloudFront
Cache	None	POC, not needed yet	Redis (embeddings + responses)
Database	None	In-memory, ephemeral	PostgreSQL (users, history)
Auth	None	POC only	OAuth2 + JWT + RBAC

5. Document Sources

All 4 sources are searched simultaneously (parallel Serper queries), not sequentially. Results are merged and deduplicated before processing.

Source	URL	What it covers	Status
Indian Kanoon	indiankanoon.org	SC, all HCs, Tribunals, 10cr+ cases	Active
LiveLaw	livelaw.in	Recent HC/SC judgements + legal news	Active
Bar & Bench	barandbench.com	HC/SC judgements + legal analysis	Active
Supreme Court	sci.gov.in	Official SC judgements	Active
SCC Online	scconline.com	Premium, best quality	Future
Manupatra	manupatra.com	Premium, comprehensive	Future
JUDIS	judis.nic.in	SC archive	Future

6. Pipeline Timings (Free Tier)

Stage	Time	Bottleneck?	How to Improve
Extract (LLM)	~2-3s	No	Already optimized (thinking disabled)
Search (Serper)	~1-2s	No	Parallel queries, fast
Scrape (10 URLs)	~4-6s	Sometimes	Increase concurrency (semaphore=5)
Chunk + Embed	~4-5s	Yes (quota)	Paid tier = no waiting; Redis cache
FAISS search	<0.1s	No	Already near-instant
Validate (LLM)	~3-5s	Sometimes	Stream response to show partial results
TOTAL	~15-20s	--	Target: <10s on paid tier with caching

7. Known Limitations (Free Tier)

Limitation	Detail	Workaround / Fix
Embedding quota	100 req/min free tier	20-chunk cap; 62s retry; paid tier
Render cold start	40s first request after 15min idle	Retry once; upgrade to paid
Chunk cap (20)	Less context for FAISS	Paid Gemini = increase to 100+
Serper 2500/month	High usage may exhaust	IK API as backup
No auth/login	Anyone with URL can use	Add Google OAuth2
No history	Each search is independent	PostgreSQL + user sessions
LLM hallucination	Case names may be inaccurate	Add citation verification step

8. Future Scope & Enterprise Roadmap

8.1 Open Source / Offline LLMs (Ollama Support)

Current system uses Gemini (cloud API). For law firms with confidentiality requirements, client data should never leave the firm's servers. Solution: Ollama.

Ollama is a tool that runs LLMs locally on any laptop or server -- completely free, completely offline. No API key, no data sent to Google/OpenAI.

```
# Current (Gemini - cloud)
client = genai.Client(api_key='AIza...')
resp = client.models.generate_content(model='gemini-2.5-flash', ...)

# Future (Ollama - local, free, offline)
# pip install ollama
import ollama
resp = ollama.chat(model='llama3.2', messages=[{...}])

# Recommended models for Ollama:
# LLM:          llama3.2 / mistral / phi-4
# Embeddings: nomic-embed-text / mxbai-embed-large
```

Feature	Gemini (Cloud)	Ollama (Local)
Cost	Pay per request (~Rs 0.01/search)	Free forever
Privacy	Data sent to Google servers	100% local, no data leak
Internet	Required	Not required after setup
Speed	Fast (Google infrastructure)	Depends on your hardware
Best for	POC, startups, small teams	Law firms, hospitals, courts
Setup	API key only	GPU/good CPU server needed

8.2 Enterprise Scale Architecture

POC (Now)	MVP (3 months)	Enterprise (6+ months)
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Streamlit UI	-> React + TypeScript	-> White-label product
Single Render server	-> Docker + CI/CD	-> Kubernetes auto-scale
Gemini only	-> Multi-model support	-> Ollama option
No DB	-> PostgreSQL	-> Multi-tenant DB
No auth	-> Google OAuth2	-> SSO + RBAC
No cache	-> Redis cache	-> Full caching layer
IK + 3 sources	-> 10+ sources	-> SCC Online + Manupatra

FAISS (memory)	-> Pinecone/Qdrant	-> Custom vector index
Manual key mgmt	-> Secret Manager	-> HSM / Vault

8.3 Additional Features for Enterprise

- Multi-language support -- Hindi, Marathi, Tamil case documents
- Case timeline analysis -- see how law evolved over time on a statute
- Judge-specific search -- find all judgements by a specific judge
- Citation network -- which cases cite each other (graph visualization)
- Bulk case processing -- upload 100 cases at once, process overnight
- API access -- integrate into existing legal practice management software
- Mobile app -- React Native app for lawyers on the go
- Offline mode -- download case embeddings, search without internet
- Audit trail -- log every search for compliance and billing
- Custom fine-tuning -- fine-tune LLM on Indian legal corpus for better accuracy

9. Quick Reference

Live URLs

Frontend (Streamlit): <https://legalmate-5yuktet3xgffj2xcbc8y5l.streamlit.app>

Backend (Render): <https://legal-search-api-y9ca.onrender.com>

GitHub Repository: https://github.com/swarajsaxena9-FI/legal_mate

Health Check: <https://legal-search-api-y9ca.onrender.com/health>

Environment Variables Required

Variable	Required	Where to Get
GEMINI_API_KEY	Yes	https://aistudio.google.com/apikey
SERPER_API_KEY	Yes (on cloud)	https://serper.dev
GOOGLE_CSE_API_KEY	Optional	Google Cloud Console
GOOGLE_CSE_ID	Optional	https://programmablesearchengine.google.com
JINA_API_KEY	Optional	https://jina.ai/reader

Local Setup Commands

```
git clone https://github.com/swarajsaxena9-FI/legal_mate.git
cd legal_mate
python -m venv .venv
.venv\Scripts\activate          # Windows
pip install -r requirements.txt
cp .env.example .env           # Fill in your API keys

# Terminal 1 - Backend
uvicorn app.main:app --reload --port 8000

# Terminal 2 - Frontend
streamlit run streamlit_app.py
```